



Using contextual data to predict risky driving events: A novel methodology from explainable artificial intelligence

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ABSTRACT

Usage-based insurance has allowed insurers to dynamically tailor insurance premiums by understanding when and how safe policyholders drive. However, telematics information can also be used to understand the driving contexts experienced by the driver within each trip (e.g., road types, weather, traffic). Since different combinations of these conditions affect exposure to accidents, this understanding introduces predictive opportunities in driving risk assessment. This paper investigates the relationships between driving context combinations and risk using a naturalistic driving dataset of 77,859 km. In particular, XGBoost and Random Forests are used to determine the predictive significance of driving contexts for near-misses, speeding and distraction events. Moreover, the most important contextual factors in predicting these risky events are identified and ranked through Shapley Additive Explanations. The results show that the driving context has significant power in predicting driving risk. Speed limit, weather temperature, wind speed, traffic conditions and road slope appear in the top ten most relevant features for most risky events. Analysing contextual feature variations and their influence on risky events showed that low-speed limits increase the predicted frequency of speeding and phone unlocking events, whereas high-speed limits decrease harsh accelerations. Low temperatures decrease the expected frequency of harsh manoeuvres, and precipitations increase harsh acceleration, harsh braking, and distraction events. Furthermore, road slope, intersections and pavement quality are the most critical factors among road layout attributes. The methodology presented in this study aims to support road safety stakeholders and insurers by providing insights to study the contextual risk factors that influence road accident frequency and driving risk.

1. Introduction

The proliferation of telematics devices has enabled motor insurance companies to enhance their actuarial models by understanding how safe and how much policyholders drive. Generally known as Usage-based Insurance (UBI), these actuarial models use dynamic driving patterns and traditional risk factors to associate policyholder profiles with claims frequency and severity. In particular, the ability to track dynamic driving information such as driver location, behaviour and distance

travelled has introduced considerable predictive gains to actuarial models (Ayuso et al., 2019; Huang & Meng, 2019; Verbelen et al., 2018). However, while driving location and behavioural information improves actuarial models, it is unclear how incorporating the latent driving context (i.e., where people drive) influences them. Such driving context encompasses the external conditions affecting the driving task, including environmental, infrastructural, and traffic aspects (e.g., weather conditions, road type and layout, traffic speed).

This study proposes a novel methodology to analyse driving

Abbreviations: ADAS, Advanced Driver Assistance; Systems GLM, Generalized Linear Models; GNSS, Global Navigation Satellite; System IRI, International Roughness Index; MPD, Mean Poisson Deviance; PAYD, Pay-as-you-drive; PHYD, Pay-how-you-drive; PWYD, Pay-where-you-drive; RMSE, Root Mean Squared Error; SHAP, Shapley Additive; Explanations UBI, Usage-based Insurance.

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exposure to hazardous events through driving context profiles collected from telematics devices. These profiles include combinations of lighting conditions, road infrastructure, layout, traffic conditions, traffic signs, and weather information. Although road safety organisations use such contextual variables to understand the causes of accidents, their application in the motor-insurance industry remains limited. To the best of the authors' knowledge, this is the first research that investigates associations between a comprehensive set of driving context combinations and their exposure to dangerous driving situations, sharing with the scientific community the collected dataset for further investigation. Moreover, it explores the most relevant contextual categories to assist the decision-making process of insurers and road safety stakeholders willing to incorporate spatial factors in their analyses.

The driving context profile adds a layer of information to telematics-based insurance. Before the emergence of telematics in the motor insurance sector, insurers had to rely on static risk factors that proxy for the inherent driving risk. These factors are generally related to vehicle characteristics (e.g., brand and model), driver demographics (e.g., place of residence) and historical claims data (Antonio and Valdez, 2012; de Jong and Heller, 2008; Denuit et al., 2007). A particular fact of these models is that they depend on claims to achieve a considerable predictive level. However, due to the inherent nature of road accidents, their frequency in an insurance portfolio is low (Lord & Mannering, 2010). With the introduction of Usage-based Insurance schemes (UBI), insurers use telematics devices to have a dynamic source of information about accident exposure, even before an accident happens (Paefgen et al., 2013). Telematics devices can detect hazardous driving situations that might have led to a road accident. Hence, in the telematics-based insurance literature, these events act as safety variables associated with road accidents (Guillen et al., 2021; Mao et al., 2021). This research aligns with previous works by using such dynamic situations as a proxy for driving risk and refers to them as risky events. In particular, these situations encompass vehicle motion anomalies that may lead to an accident, also known as near-misses (e.g., harsh acceleration, braking, and swerving), and hazardous driving patterns like phone use while driving and speeding.

The first contribution of this paper is the investigation of near-misses and hazardous driving patterns related to distraction and speeding using a comprehensive set of driving context combinations, including road environment, road infrastructure and topology, traffic conditions, road signs, and weather and lighting conditions. Furthermore, the analytic methodology presented in this paper serves to understand and interpret the predictive effects of driving contexts and their ranking of importance in the computation of risk. Although exposure to near-miss events has been used in the insurance telematics literature (Guillen et al., 2021; Guo et al., 2010; Sun et al., 2021), little focus has been put on the relationship between these events and the driving context in which they happen. In the Road Safety domain, previous studies analysed the relevance of different contextual factors on accident exposure. However, no specific research addresses the analysis of near-misses, speeding, and distraction events using a set of 18 risk factors from the aforementioned contextual categories. Moreover, previous studies in the Road Safety literature have already applied Explainable Artificial Intelligence techniques to interpret risk predictions, rank risk factors, and analyse risk factors' effects on model outputs. Nevertheless, none of them analysed telematics data collected from vehicles and their relationship with dangerous events such as near-misses, speeding, and distraction. Instead, they examine data from accident statistics with risk factors related to their geographical location, for instance, the road network, road facilities, and the region's demographics.

In this research article, we analyse a naturalistic driving dataset to answer the following research questions: (i) does the driving context carry predictive power for measuring driving risk?; (ii) what are the most relevant contextual features to evaluate risk exposure?; (iii) how do contextual factors from five different contextual categories influence the exposure to near-misses and hazardous patterns? Studying

naturalistic driving data allows thorough analyses of the leading causes of dangerous traffic events, helping insurers and road safety stakeholders develop data-driven policies. However, publicly available naturalistic driving data is often difficult to obtain (Masello et al., 2021). We counteract this universal barrier and foster innovation in the motor insurance industry by contributing a publicly available dataset and encouraging research teams to investigate the relationship between driving context and risk and its added value to their risk models.

This research presents the basis of a potential new scheme of Usage-based Insurance – Pay-where-you-drive (PWYD). Studying driver behaviour through driving context introduces predictive opportunities in driving risk assessment as the exposure to accidents varies according to contextual factors such as the road type or the weather conditions (UK Department for Transport, 2020; Zhu et al., 2017). However, although tracking driving patterns introduce ratemaking benefits for insurers, they also carry privacy concerns (Goldfarb & Tucker, 2012; McDonnell et al., 2021). Thus, context profiles might also provide insurers with an alternative approach to approximate policyholders' exposure to accidents without measuring driving habits at a high sampling rate and posing privacy concerns. This study also assists road safety stakeholders willing to analyse contextual dependencies and their inherent risk and derive data-driven policies backed by explainable machine learning techniques.

The paper is organised as follows. Section 2 presents previous studies in driving exposure and behaviour, near-miss events, driving context analysis, modelling approaches to risk assessment, and explainable artificial intelligence in risk analysis. Section 3 introduces the dataset and develops the notion of the driving context profile, detailing statistical observations on the dataset. Section 4 elaborates on the methods used within this paper, covering the two regression techniques – XGBoost and Random Forest –, the feature transformation used to train these models, and the method used to extract feature importance – Shapley Additive Explanations. Section 5 presents the results of modelling the relationship between driving context combinations and near-misses, speeding, and distraction events. Moreover, Section 5 analyses the effects of contextual variables on models' output and presents a ranking of feature importance. The paper concludes with the main conclusions and lines for future work in Section 6.

2. Background

2.1. Driving exposure and behaviour

Driving exposure and behaviour are widely studied topics in motor insurance and road safety research, enabling two central schemes for the insurance sector – Pay-as-you-drive (PAYD) and Pay-how-you-drive (PHYD). These schemes leverage different levels of driving habits to tailor the insurance premiums to the policyholders' risk (Baecke and Bocca, 2017). Pay-as-you-drive schemes measure exposure to traffic accidents through distance and frequent road types (Tselentis et al., 2017). For instance, Paefgen et al. (2013) studied risk exposure using the travelled distance, time of day, day of the week, road types and driving speed and found that distance was the most significant predictor, as Vickrey (1968) suggested. Guillen et al. (2019a) suggested that large distances should correspond to higher premiums, although moderated by the driving experience factor. In contrast to PAYD, PHYD uses driver behaviour to segment policyholders according to their risk (Handel et al., 2014; Tselentis et al., 2017). Such a segmentation carries a more comprehensive layer of information that reflects different drivers' risk appetites, helping insurers enhance the accuracy of their actuarial models (Ayuso et al., 2019; Bian et al., 2018).

Studies of driving behaviour are generally undertaken using naturalistic driving datasets, as Ahmed et al. (2022) mentioned in a systematic review. Several driving patterns serve to measure or classify such driving profiles, from vehicle dynamics like speeding patterns and aggressive manoeuvres (e.g., harsh braking, swerve) to driver-centric

Table 1

Comparison of previous studies using driving context data as risk factors for modelling driving risk.

Study	Contextual risk factors				
	Road environment	Road infrastructure and topology	Traffic conditions	Traffic signs	Weather and lighting conditions
Abellán et al. (2013)	Rural roads	Lane width, shoulder type, pavement width, lane markings, shoulder width, road dividers.	–	–	Weather conditions (good, heavy rain, light rain, other), lighting conditions (daylight, dusk, nighttime)
Guillen et al. (2019a)	Speed limit and urban roads	–	–	–	Night driving
Guillen et al. (2019b)	Speed limit and urban roads	–	–	–	Night driving
Hu et al. (2019)	Speed limit and road types (freeways, arterials, local roads)	–	Traffic volume, traffic speed, peak hours	–	Night driving
Jun et al. (2011)	Speed limit and road types (freeways, arterials, local roads)	–	Peak hours	–	–
Ma et al. (2018)	Speed limit	–	Traffic volume, traffic speed, peak hours	–	–
Mase et al. (2020)	Speed limit	–	–	–	Rainfall and daylight
Montella & Imbriani (2015)	Rural motorways	Segment length, radius, deflection angle, superelevation, vertical curves, sight distance, shoulder width.	Traffic volume	–	Weather conditions (clear, rainy, wet, non-wet), lighting conditions (daylight, nighttime)
Paefgen et al. (2013)	Road types (urban, highway, extra-urban)	–	Time of day	–	–
Saeed et al. (2019)	Highways	Segment length, lane width, shoulder width, road quality (IRI), road divider, number of lanes, tangent segments	Traffic volume	–	–
Wang et al. (2015)	Road types (Structured, normal, hybrid, rural)	Intersections, road dividers	–	–	Weather conditions (sunny, cloudy, others), lighting conditions (daylight, dusk)
Yang et al. (2019)	Urban roads, land use (commercial, residential, park), demographics	Intersections, road length, bus stops	Traffic volumes, peak hours	–	Daylight and night driving
Zhu et al. (2017)	Speed limit, road types (freeway, artery, ramp, highway, minor)	–	Traffic speed	–	–
Our research article	Speed limit, road types (urban, rural, motorways)	Distance to intersection, road quality (IRI), number of lanes, road bend, road slope, tunnel	Geolocated traffic category	No overtaking area, traffic lights, yield signs	Lighting conditions (daylight, nighttime), precipitation, temperature, visibility, weather conditions (clear, cloudy, fog, rain, snow/ice), wind speed

factors such as distraction and impairment. Speeding is among the most studied factors associated with exposure to accidents (Ayuso et al., 2010; Ellison et al., 2015; Jun et al., 2011) and their expected severity (Shannon et al., 2018). Aggressive manoeuvres represent other widely investigated factors due to their correlation to the driving risk caused by significant deviations in kinematics values (Ryan et al., 2019). Several studies found that harsh acceleration and braking positively correlate with crash frequency (af Wählberg, 2004; Ma et al., 2018; Stipanovic et al., 2018). Similarly, Klauer et al. (2009) showed that risky drivers exhibit more sudden braking, harsh acceleration and swerve manoeuvres than safe drivers. Exploiting that both speed and acceleration profiles are associated with risk, Gao et al. (2019) proposed a claims frequency model using speed-acceleration profiles as the only telematics attribute. Notably, these profiles are extracted through speed-acceleration heatmaps, which also require a low volume of data transfer that encapsulates driving patterns, although at the expense of losing information from time-series data.

Another significant factor in the analysis of driver behaviour is driver distraction. There is evidence that driver distraction is among the leading factors in road accidents, resulting in 9.7% and 7.1% of the road fatalities in the United States and the United Kingdom in 2019, respectively (NHTSA, 2020; UK Department for Transport, 2021). Several studies investigated the risk induced by this factor using naturalistic driving data (Simmons et al., 2016; Singh & Kathuria, 2021). Klauer et al. (2006) analysed the relationships between driver inattention and risk using one of the most extensive data collection campaigns in the

United States, the *100-Car* dataset (Dingus et al., 2006).

The authors found a three-fold increase in the driving risk for drivers engaging in visually or manually complex tasks. Most importantly, they found that distraction in particular driving contexts can lead to higher risk (e.g., distraction in intersections, slippery roads, and congested traffic conditions). Other studies analysed driver distraction with a significant naturalistic dataset in Europe, the UDRIVE project. Hibberd et al. (2020) observed a positive correlation between aggressive drivers and phone usage, and Carsten et al. (2017) found that drivers tend to engage in distracting activities for 10% of the driving time. Notably, phone use was the most frequent factor, at 4% of driving time. Using behavioural data from a young novice drivers survey, Jannusch et al. (2021) noted that high-risk driving behaviours such as speeding are more likely to be observed in drivers who engage in phone calls while driving.

2.2. Near-miss events

A drawback of traditional actuarial models is the high dependency on claims occurrence, which is infrequent due to the nature of traffic accidents (Lord & Mannering, 2010). Because driving behaviour is highly associated with risk, an alternative line of inquiry uses dangerous events that might have led to accidents as a proxy for claims (Guillen et al., 2021; Guo et al., 2010; Sun et al., 2021). These events, also called near-misses, allow overcoming modelling issues due to infrequent claims while approximating the drivers' exposure to accidents. The

higher frequency of near-misses also introduces opportunities for dynamic premiums based on the driving patterns. For instance, Guillen et al. (2021) posited a method to update the policyholder's premium weekly based on the number of near-misses, leveraging the associations found between claims frequency and near-misses.

Several studies emphasised the importance of near-misses for UBI. Guo et al. (2010) suggested using near-misses as a risk indicator when there is a significant absence of crash data for statistical analysis. If both variables are available, Guillen et al. (2019b) proposed utilising a combination of them to improve claims modelling performance. Sun et al. (2021) grouped drivers into risk levels according to four near-misses: speeding, high-speed braking, harsh acceleration, and harsh braking. Ma et al. (2018) found that harsh braking and starts are highly correlated with accident involvement. Seacrist et al. (2020) performed a similar risk categorisation, analysing the main characteristics of 4,818 instances of near-misses across age groups. The authors found that young drivers experience more rear-end and lane departure near-misses than adult and older drivers. Furthermore, mobile phone distractions were among the top distractions before near-misses occurred. The relevance of such distractions was also emphasised by Guillen et al. (2021), who introduced premium penalisation for active mobile phone use.

The detection of near-misses is typically performed using thresholds for longitudinal and lateral acceleration. These refer to extreme values associated with manoeuvres required to avoid an accident. In previous studies, harsh acceleration and braking are triggered with values of approximately 6 and -6 m/s^2 (Guillen et al., 2021; Klauer et al., 2006; Lee et al., 2011; Seacrist et al., 2020; Sun et al., 2021). These values are associated with sudden movements observed in whiplash events (Hynes & Dickey, 2008). Likewise, swerving manoeuvres are extreme lateral movements of $\sim 7.5 \text{ m/s}^2$, resembling the no-sliding condition of Wahlström et al. (2015) (Davoodi et al., 2021; Guillen et al., 2021; Klauer et al., 2006; Lee et al., 2011).

2.3. The influence of the driving context

Traditional actuarial models and UBI approaches complement each other to obtain the best predictive power (Gao et al., 2022a). The reason is that classical factors help understand the circumstances under which the driving patterns are captured (Gao et al., 2022). Another factor contributing to such an understanding is the driving context, which captures the spatial context of driving habits and leads to enhanced performance in risk assessment models (Zhu et al., 2017). Table 1 compares previous works that studied contextual factors and their association with driving risk, and highlights the contribution of this research article with respect to the driving context. With the aim to provide a comprehensive assessment, the table includes studies that use the notion of driving risk according to crash frequency, near-miss frequency, and safety performance functions. The table performs the comparisons using the following risk factor categories: (i) road environment, (ii) road infrastructure and topology, (iii) traffic conditions, (iv) road signs, and (v) weather and lighting conditions.

The road environment (e.g., road types and speed limits) are among the most studied factors that influence driving risk. Based on naturalistic driving data and crash history surveys, authors Jun et al. (2011) and Ma et al. (2018) found that exceeding the speed limit is highly correlated with accident risk. A similar observation was found by Guillen, Nielsen, Ayuso, et al. (2019), who also posited that driving on urban roads is associated with high exposure to accidents.

The road infrastructure and topology are principal factors when designing road safety performance functions due to their link to accidents. Montella & Imbriani (2015) studied the effects of the road infrastructure on the safety performance of motorways by measuring the relationship between geometric design and road accident frequency. The results showed that inconsistencies in the road geometric design (e.g., in road quality, slopes, and curve alignment) increase accident risk on

motorways. The road geometry and quality was also found to be a main factor for highway design aiming at reducing accidents in Saeed et al. (2019). Traffic signs comprise another piece of information when examining driving risk. According to the United Kingdom's road safety reports for 2019, disobeying traffic lights, priority and non-overtaking signals represented 0.56%, 1.41%, and 1.83% of the contributing factors for fatal accidents (UK Department for Transport, 2021).

Regarding traffic conditions, Ma et al. (2018) found that adding risk factors related to traffic volume, traffic speed, and the legal speed limit improved the accident frequency prediction. In particular, road congestion and the speed relative to neighbouring vehicles are relevant predictors of driving risk. Similar conclusions were found by Hu et al. (2019) analysing risk factors in different traffic and road conditions, and using the same dataset containing telematics and crash history. Zhu et al. (2017) found that driving at speeds faster than the traffic speed at ramps increases the driving risk. Based on the fact that different traffic conditions affect road safety throughout times of day, Yang et al. (2019) investigated driving risk using several time ranges within Manhattan census tracts. The authors used naturalistic driving data, governmental data related to crashes, traffic volumes, land use, and demographic data for the studied region. The results showed time-dependent risk areas, highlighting that risk hotspots vary according to times of day.

Environmental factors are other widely used variables to measure driving risk. Adverse environmental factors increase the driving risk, as shown by Abellán et al. (2013), who studied adverse weather and lighting conditions and their link with crash severity. Mase et al. (2020) analysed two weather factors – rainfall and daylight conditions – and their influence on driving risk for heavy goods vehicles. The authors found that daylight driving is positively associated with speeding whereas rainfall is negatively associated with speeding, harsh braking, and harsh cornering. A limitation of this study is that the authors linked telematics data with aggregated weather data for the whole region instead of the weather conditions based on trip time and geolocation, like in our research article. Guillen et al. (2019b) found that night driving is associated with a lower probability of cornering events while driving on urban roads with a higher likelihood of harsh braking. Wang et al. (2015) integrated driving behaviour, vehicle features and contextual variables for a risk assessment method based on near-misses. However, the impact of the contextual factors on driving risk was not comprehensively analysed.

While the aforementioned studies showed the relevance of driving contexts on road safety, their analyses were limited to isolated risk factors (e.g., speed limit, road type, traffic conditions). Furthermore, there is no specific research about how driving under several contextual combinations of the five studied categories influences exposure to dangerous events. To fill this gap, this research article analyses the occurrence of near-misses, speeding, and distraction events across a comprehensive set of 18 contextual features. Moreover, the article analyses feature rankings showing the most relevant variables for each risk event, together with the effects of risk factors on the model output.

2.4. Modelling techniques in driving risk assessment

Traditionally, actuarial models use Generalized linear models (GLMs) to infer risk and predict claims and near-misses frequency and severity (Denuit et al., 2007; Lemaire et al., 2016; Sun et al., 2021). Other lines of research approached the risk assessment problem using machine learning techniques (Wen et al., 2021a). For instance, Paefgen et al. (2013) studied PAYD with logistic regression, neural networks, and decision trees. The authors observed that logistic regression was the most suitable due to its interpretability aspects required by insurance stakeholders. A negative aspect of this technique is that it might present difficulties in learning complex non-linear relationships (Moro et al., 2014). Neural networks generally overcome this issue at the expense of decreasing transparency and interpretability (Baecke and Bocca, 2017). To reduce this tradeoff, Gao et al. (2022) combined the predictive gains

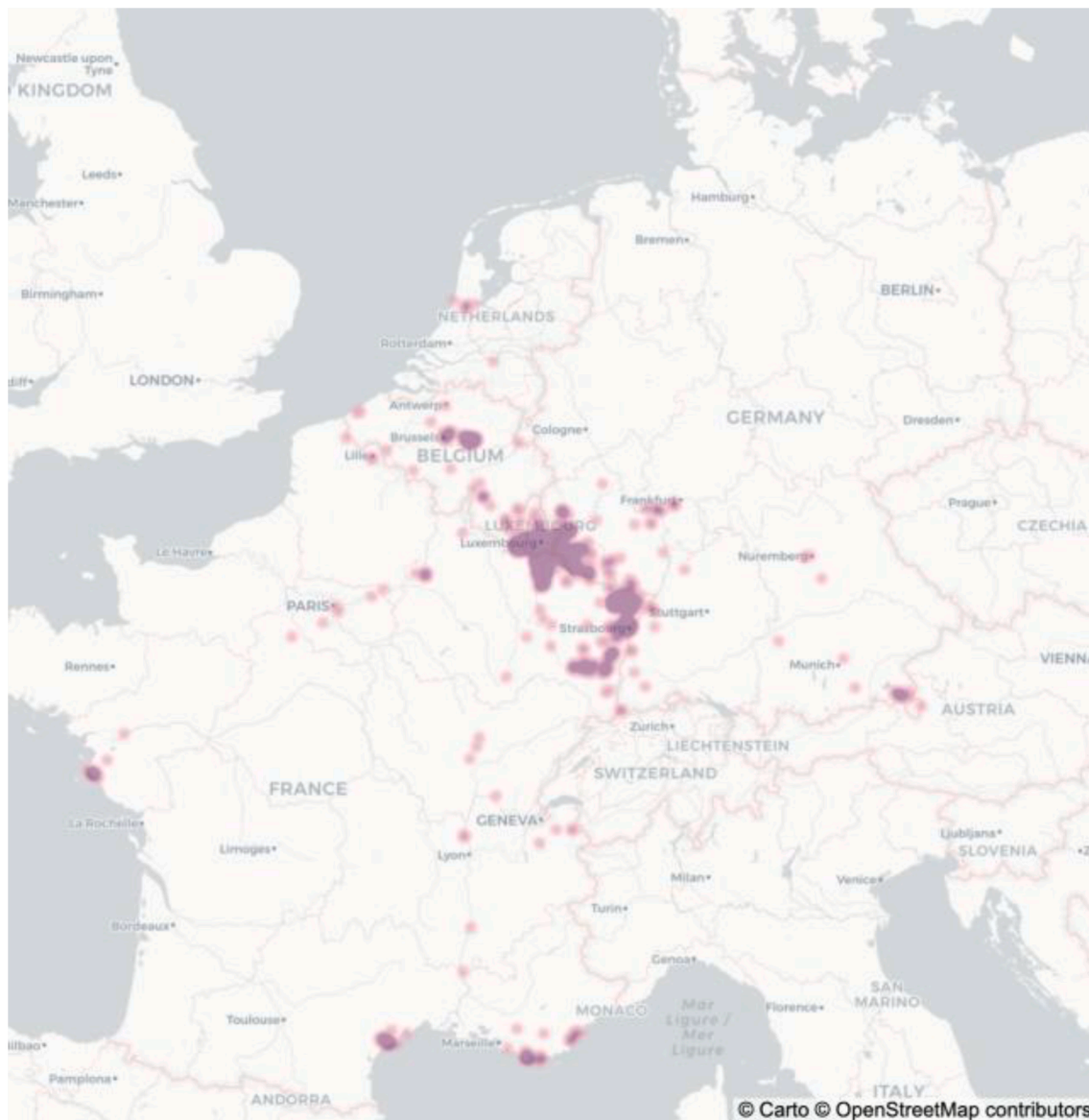


Fig. 1. Coordinates of the dataset's trips. The paper considers 3,220 trips performed in the region of Belgium, France, Germany, Luxembourg, and the Netherlands. A high density of trips can be observed around Luxembourg and France's Grand Est region.

of neural networks with the interpretability features of GLMs. The authors introduced a driver behaviour component obtained through neural networks as an additional attribute in a GLM model, and the results showed increased performance in claims frequency modelling.

Due to the potential learning abilities and transparency features, tree-based machine learning methods are other widely used techniques in the risk assessment and credit scoring literature (Lessmann et al., 2015). Henckaerts et al. (2021) modelled claims frequency and severity with ensembles of decision trees (i.e., random forest and boosted trees) and observed that such techniques outperformed the traditional GLMs. Similarly, Huang & Meng (2019) found that such ensembles presented superior accuracy and robustness in risk classification problems, with XGBoost showing better performance than Random Forest. The performance of XGBoost in risk assessment was also analysed by Pesantez-Narvaez et al. (2019) in a claims prediction problem, positing that while XGBoost might overcome logistic regression performance, in imbalanced datasets, it requires a considerable fine-tuning process.

2.5. Explainable artificial intelligence for risk analysis

With recent advances in computer science, explainable artificial intelligence (Explainable AI) has become a principal enabler of machine learning in domains that require a clear understanding of the model predictions, such as road safety and insurance. Consequently, several explainability techniques have been developed. Focusing on crash frequency modelling, Wen et al. (2022) compared the strengths and weaknesses of such techniques. The authors evaluated two types of explainability methods: model-based – Classification and Regression Tree (CART), Multivariate Adaptive Regression Splines (MARS) – and post hoc – Local Interpretable Model-agnostic Explanations (LIME), Local Sensitivity Analysis (LSA), Partial Dependence Plots (PDP), Global Sensitivity Analysis (GSA), and Shapley Additive Explanations (SHAP). The results showed that SHAP was the best technique to capture the correlation among risk factors without constraining the model complexity and to visualise relationships between the model prediction and its predictors.

The interpretability features of SHAP, with their profound basis in mathematical properties, have made this technique a helpful resource

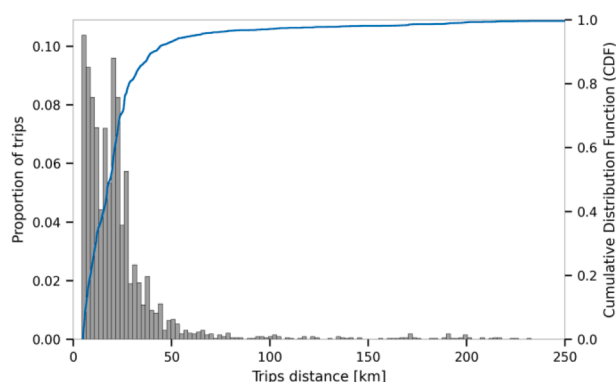


Fig. 2. Distribution of distance of the 3,220 dataset's trips (mean = 24.9 km, median = 18.7 km, and standard deviation = 33.2 km). The majority of the trips (95%) travelled less than 60 km.

Table 2

Distribution of the number of trips and proportion of kilometres tackled by months. The distance travelled is distributed uniformly across months despite drivers being tracked for 88 days on average (std = 60 days).

Month	Number of trips	Proportion of total distance [%]
July 2021	291	11.7
August 2021	415	12.7
September 2021	465	12.6
October 2021	509	16.3
November 2021	450	14.4
December 2021	441	13.2
January 2022	360	10.7
February 2022	289	8.4

for an increasing number of research teams. Mihaita et al. (2019) used such a technique to analyse road safety variables' importance in predicting accident duration. Wen et al. (2021) used SHAP to study the influence of several risk factors on the frequency of different types of crashes (e.g., rear-end collisions). In particular, the authors analysed individual and interaction effects on crash frequency predictions. The importance of each risk factor was found to vary according to the accident type; thus, the authors posited that claims modelling should be disaggregated across accident types. Chang et al. (2022) also analysed interaction effects between road safety attributes. The authors investigated the non-linear relationships between risk factors and the predicted likelihood of fatal pedestrian accidents. SHAP enabled the discovery of insightful interactions between risk factors and model predictions; for example, the probability of a deadly pedestrian accident increases when dense residential areas are combined with low-density intersections. Similarly, Yang et al. (2021) analysed traffic accidents involving heavy goods vehicles. Examining SHAP plots, the authors identified non-linear effects between the risk factors and accident predictions (e.g., the likelihood of possible injury crashes increases in areas with higher employment density).

Interaction effects are often studied together with feature importance rankings. For instance, Parsa et al. (2020) leveraged SHAP to interpret traffic accident occurrences in expressways through interaction effects and feature importance. The authors analysed feature importance in accident detection and found that significant differences in speed between expressways' traffic measurement sites were the most relevant factor. Applying a similar analysis, but focusing on intersection accidents, J. Hu et al. (2020) used SHAP to rank risk factors according to their contribution to the predicted likelihood of accidents. Particularly, this study uses force plots to illustrate feature contributions to the model predictions at three levels: high risk, medium risk, and low risk.

While the aforementioned studies introduced significant contributions in the road safety domain through the application of SHAP, none of

them analysed telematics data collected from vehicles and their relationship with dangerous events such as near-misses. Instead, they examine data from accident statistics with risk factors related to their geographical location; for instance, the road network (e.g., road type, geometry and traffic counters), road facilities (e.g., pedestrian crossing area, and land use), and demographics. This article fills this research gap by applying SHAP to investigate the likelihood of dangerous driving events (i.e., near-misses, speeding, and distraction) using contextual risk factors collected from naturalistic driving data.

3. Dataset

3.1. Data collection

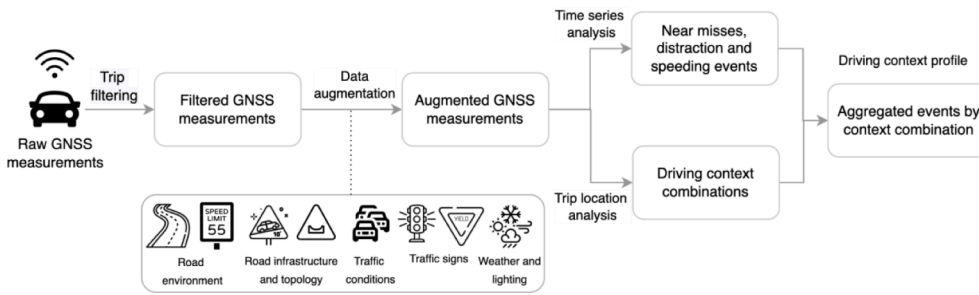
This paper uses anonymous telematics data collected from 32 drivers between July 1st, 2021, and February 20th, 2022, in the geographic regions of Belgium, France, Germany, Luxembourg, and the Netherlands. Fig. 1 shows the coordinates of the dataset's trips, where it can be observed most trips were performed around Luxembourg and France's Grand Est region. In total, 3,220 trips with at least five kilometres were considered, giving 77,859 km of total distance travelled with a mean of 24.9 km (std = 33.2 km) per trip. The distribution of trip distances is illustrated in Fig. 2, showing that 95% of the trips have less than 60 km. Although drivers were tracked for 88 days on average (std = 60 days), the distance covered is distributed uniformly across months, as shown in Table 2.

The trips were collected using the *Motion-S Trip Recorder* mobile application that collects Global Navigation Satellite Systems (GNSS) samples at 1 Hz (Motion-S, 2022a). Each trip is processed using Motion-S's pipeline, consisting of three main layers—path filtering, augmentation, and profiling—illustrated in Fig. 3. The trip path filtering corrects GNSS measurement errors using a map-matching approach where all deviations from the path are matched to the closest part of the road, smoothing the GNSS data sequence. Internally, the platform uses HERE's Route Matching V8 process that corrects GNSS errors with the most likely driven route based on the trajectory and considers potential GNSS inaccuracies (HERE, 2022). The process takes raw GNSS measurements and generates a path by matching GNSS traces to the nearest road and filtering out outliers. GNSS data are augmented with contextual information about the trip path using the categories described in Table 3 (Motion-S, 2022b). Thus, each trip location contains contextual features combined with the vehicle's motion data. The third phase uses the augmented GNSS data to obtain a trip profile according to the study domain of interest; for this paper, this consists of the driving context profile.

The dataset presented in this research aligns with previous studies mentioned in Section 2, where dangerous driving events are used as a proxy for exposure to road accidents. Research in insurance-based telematics typically uses such variables to perform risk segmentation in the absence of accidents. The collected dataset is unique in the way that it associates such risky events with a comprehensive set of contextual factors used in accident report analyses (Amini et al., 2022). Moreover, while publicly available naturalistic driving datasets exist in the United States, like the 100-Car (Dingus et al., 2006), there is an absence of such datasets in the geographical region of this study. By making this dataset publicly available, researchers have access to naturalistic driving data in Europe to study associations between the driving context and risky events.

3.2. Driving context profile

The driving context profile analyses driving motion dynamics looking for near-misses, distractions, and speeding events by driving context combinations. It represents the exposure to dangerous circumstances under each observed set of contextual features, composed of the variables described in Table 4. Numerical features such as speed limit and

**Table 3**

Contextual categories used to augment GNSS data in the *Motion-S* platform. This information is used to extend trip location data in the “Data augmentation” process in Fig. 3.

Contextual Category	Description
Environment	High-level information about the driving environment, including road type and speed limit.
Road infrastructure and topology	Infrastructural and topological information of the road, such as slopes, curvature and road quality (measured with the International Roughness Index (IRI)).
Traffic conditions	Average traffic speed in driven areas at the time of recording.
Traffic signs	Legal traffic signs along the trip (e.g., traffic lights, yield, and no-overtaking signs).
Weather and lighting conditions	Environmental conditions of the trip, including precipitation, temperature, visibility, wind speed, sky information, and lighting conditions, among others.

road slopes are discretized to limit the number of context combinations. Therefore, the 18 contextual attributes are either categorical or binary variables. Road quality is measured using the International Roughness Index (IRI), where lower values represent better road profiles.

Table 5 describes the target events of the analysis, representing six risky driving behaviour events. These dangerous occurrences are extracted through time series analysis of the driving motion dynamics and mobile phone interactions – cornering, harsh acceleration, harsh braking, phone unlocking, phone call proportion, and speeding proportion. As presented in Section 2.2, near-miss events are detected using the acceleration and direction patterns of the vehicle. They include the detection of harsh acceleration, harsh braking and cornering events. This paper applies the thresholds used in the studies mentioned in Section 2.2, which are 6 and -6 m/s^2 for acceleration and braking events and lateral acceleration of 7.5 m/s^2 in curves for cornering manoeuvres. Speeding events are measured as the proportion of distance driven above the legal speed limit. Likewise, phone call events represent the proportion of distance driven while on a call. As a result, every trip results in a driving context profile that aggregates six risky events by groups of contextual features. The distance driven in each group is used to account for the driving exposure, which allows performing the following step – an aggregation of multiple driving context profiles.

3.3. Driving context profiles aggregation

This section details the dataset used in this paper, which is the aggregation of all context profiles obtained from the telematics data described above. Fig. 4 illustrates such an aggregation where trip context profiles are grouped by the observed contextual combinations. The target risky events are obtained through two operations – a weighted average of count events (cornering, harsh acceleration, harsh braking, phone unlocking) and a weighted average of ratio events (speeding proportion and phone call proportion) using the distance travelled in each context as the weighting factor. The use of weighted averages intends to eliminate potential biases caused by anomalous

Table 4

Contextual attributes of the driving context profile. All variables are either categorical or binary since numerical features were discretized into bins to control the number of feature combinations. Abbreviations: inf (infinite), IRI (International Roughness Index).

Contextual Category	Attribute	Range of values
Environment	Speed limit [km/h]	(0, 30], (30, 50], (50, 70], (70, 90], (90, 110], (110, 130], (130, inf)
	Road type	Motorway, Urban, Rural
	Intersection (distance to intersection less than 30 m)	True, False
	Road quality (IRI)	[0, 2], (2, 4], (4, 6], (6, 8], (8, inf)
Road infrastructure and topology	Lane category	One lane, Two or three lanes, Four or more lanes
	Road bend	True, False
	Road slope [°]	(-inf, -4], (-4, -2], (-2, -0.5], (-0.5, 0.5], (0.5, 2], (2, 4], (4, inf)
	Tunnel	True, False
Traffic conditions	Traffic category	Congested, moderate traffic, no traffic
Traffic signs	Presence of no-overtaking sign	True, False
	Presence of traffic lights	True, False
	Presence of yield sign	True, False
Weather and lighting information	Lighting conditions	Day, Night
	Precipitation [mm/h]	[0, 0.5], (0.5, 1.5], (1.5, 2.5], (2.5, inf)
	Temperature [°C]	(-inf, -10], (-10, 0], (0, 10], (10, 20], (20, 30], (30, 40], (40, inf)
	Visibility [m]	(0, 2,500], (2,500, 5,000], (5,000, 7,500], (7,500, 10,000]
	Weather conditions	Clear, Cloudy, Fog, Rain, Snow or ice
	Wind speed [km/h]	[0, 10], (10, 20], (20, 30], (30, inf)

observations. Thus, the resulting dataset represents an aggregated exposure to dangerous events by contextual combinations observed by anonymous drivers. Fig. 5 presents an example of such an aggregation using two trips.

The resulting dataset consists of 145,842 unique context combinations with 1,985 harsh brakings, 819 harsh accelerations, 2,683 cornering events, 1,628 phone unlockings, and 15% and 4% speeding proportion and phone calls on average. Notably, the percentage of phone calls proportion aligns with the findings of the distraction study performed on the UDRIVE project (Carsten et al., 2017). Fig. 6 details the feature distribution of categorical and binary variables aforementioned in Table 3. There is a higher proportion of urban and rural roads and relatively low-speed limits because such contexts present more variability in contextual combinations. For example, while motorways

Table 5

Target risky events. Four count events (cornering, harsh acceleration, harsh braking, and phone unlocking) and two ratio events (phone call proportion and speeding proportion) are analysed using the thresholds posited by previous studies (Section 2.2).

Risky events	Description
Cornering	Number of non-consecutive lateral accelerations exceeding 7.5 m/s ² in curves over distance travelled.
Harsh acceleration	Number of non-consecutive accelerations exceeding 6 m/s ² over distance travelled.
Harsh braking	Number of non-consecutive brakings with magnitudes less than -6 m/s ² over distance travelled. Phone call proportion
Phone unlocking	Proportion of kilometres driven while making phone calls. Number of distraction events caused by unlocking the phone over distance travelled. Speeding proportion Proportion of kilometres driven exceeding the legal speed limit.

have good road quality (IRI), rural and urban roads explain more variance in most cases, reaching poor levels of IRI values. Moreover, these roads are more exposed to intersections and traffic lights, which increases the number of contextual combinations. Weather-related features tend to show moderate to low temperatures with cloudy and rain conditions. This fact was expected due to the geographical region of the data collection (Fig. 1). Furthermore, as some trips were collected in Germany, some speed limits fall into the (130 km/h, inf) speed range, referring to motorways with no constraints in this attribute.

Fig. 7 illustrates the most relevant associations between features and risky events. The point-plots show estimates of the central tendency (in this case, the mean) of risky events by contextual features. It can be observed that speed limits and road types present considerable correlations in most of them, followed by weather- and traffic-related variables. For example, there is 8% speeding in the context combinations involving urban roads rather than 12% in rural areas. Speeding proportion is the risky event with the most pronounced correlation with both speed limits and traffic categories, being inversely proportional to an increase in speed limits and traffic jams (i.e., speeding is most frequent in free-flow conditions). Contrastingly, harsh brakings and accelerations increase with high-speed limits and motorways. An unexpected correlation is observed between phone call proportion and adverse weather conditions. The people involved in this driving sample tend to engage in phone calls in cold and low visibility weather. While the reasons are unknown, the limited number of such adverse contexts and driving sample population might bias these correlations. Further research is encouraged to study this relationship. Aiming at fostering scientific collaboration and reproducibility, the authors make the dataset publicly available under the *Creative Commons Attribution-NonCommercial 4.0 International* licence (Creative Commons, 2020).¹

4. Methods

The paper aims to model the relationship between observed driving context combinations and the frequency of the target risky events mentioned in Table 4, highlighting the most significant features for each of them. As mentioned earlier, risky events refer to count events and ratio events and, therefore, have different assumptions about their distributions. This section describes the modelling techniques utilised for both of them, the feature transformation to fit the models, and the feature importance technique used for obtaining a ranking of the most relevant features for each risky event.

4.1. Regression models

The goal consists of finding six model hypotheses $h(x_j)$ that, given a

context combination x_j , predict the frequency of the six studied risky events for every context combination j . Since the exposure to events is not the same for all driving contexts because of differences in distances covered, the dependent variable for count events is set to the density of events per distance (i.e., number of type i events over the total distance in context combination j). This transformation does not apply for ratio events because the distance is already considered when computing the proportion of distance with speeding and distraction (Fig. 4). Thus, the dependent variable for ratio events is a real value between 0 and 1, indicating the modelled proportion p_i . It is worth noting that since context combinations with more distance represent more frequent contexts, the model training phase sets sample weights proportional to the distance. Equations 1 and 2 illustrate these approaches for modelling count and ratio events, respectively.

$$h(x_j) = \frac{\text{events}_i}{\text{km}}, \forall i \in \{\text{cornering, harsh acceleration, harsh braking, phone unlocking}\} \quad (1)$$

$$h(x_j) = p_i, \forall i \in \{\text{speeding proportion, phone call proportion}\} \quad (2)$$

As previously presented in Section 2.4, previous works applied several modelling techniques for risk analysis. This research article analyses the performance of such techniques to model the target research hypotheses, namely Random Forest, XGBoost, neural networks, and GLM. Random Forest and XGBoost are examined with a deeper focus due to their performance posited by previous works in similar study domains.

Random Forest is an ensemble learning algorithm that aggregates multiple Decision Trees' predictions (Breiman, 2001). The algorithm is typically trained via the Bootstrap Aggregating method (bagging) with a predefined number of individual decision trees that model a classification or regression problem. By averaging individual predictions, Random Forest learns the underlying relationship in the data, reducing the training set's overfitting (i.e., reducing the model variance). A particular characteristic is that it searches for the best feature among a random subset of the feature space, adding randomness when growing trees and, therefore, reducing the correlation between individual trees and the model variance.

This paper uses Python Scikit-Learn's implementation of Random Forest Regressor, which allows the selection of three different criteria to measure the quality of each split—*Squared error*, *Absolute error*, and *Poisson* (Pedregosa et al., 2011). As count events are assumed to follow a Poisson distribution, the Poisson criterion is selected for these event types. This criterion uses the reduction in Poisson deviance to find splits. In contrast, ratio events are modelled using the squared error, which utilises the split variance reduction.

The second regression technique used in this paper is XGBoost, an optimised implementation of Gradient Tree Boosting (Chen and Guestrin, 2016). This ensemble algorithm is a hypothesis boosting method that combines several weak models to develop a strong one. Its predictive power comes from sequentially fitting a new weak learner to the residual errors made by the previous one. Thus, the final prediction is determined by the sum of the individual predictions. For example, let $h(x_j)$ be the model hypothesis that predicts the density of harsh brakings in the context combination j , then if the algorithm uses three estimators, the final prediction is given by $h(x_j) = h_0(x_j) + h_1(x_j) + h_2(x_j)$. More formally, the method tries to minimise the cost function L shown by Equation (3), where l denotes a differentiable convex loss function based on the observed and predicted value; $\hat{y}_j^{(t)}$ represents the prediction to the j sample instance at the t^{th} step of the algorithm; n is the number of samples; f_t denotes the prediction given by the t^{th} learner; Ω is a regularisation term that penalises the complexity of the learner to avoid overfitting.

¹ The dataset can be requested at <https://forms.gle/C5hhfE9dbrs8nRt67>.

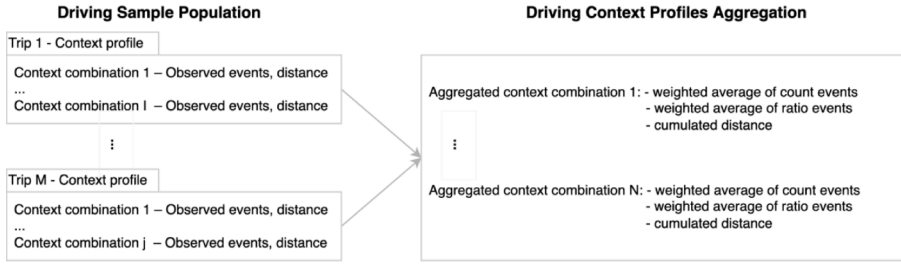


Fig. 4. Driving context profiles aggregation. The driving context profiles, extracted through the trip processing described in Section 3.2, are aggregated using a weighted average of count events (cornering, harsh acceleration, harsh braking, and phone unlockings) and weighted average of ratio events (speeding proportion and phone calls) with distance driven by context as the weighting factor. The variable M refers to the number of trips considered in the driving sample population; l and j refer to the number of distinct context combinations in trips 1 and M , respectively; N refers to the number of distinct context combinations among all trips.

$$L^t = \sum_{j=1}^n l \left(y_j, \hat{y}_j^{(t-1)} + f_t(x_j) \right) + \Omega(f_t) \quad (3)$$

This paper utilises XGBoost through the Python library XGBoost (XGBoost developers, 2021), which provides a wide range of learning objectives, including mean squared error for regression problems and Poisson for count data. As regards neural networks and GLM, the paper uses their respective Scikit-Learn's implementations: Multi-layer Perceptron Regressor (Scikit-Learn, 2022d), and Poisson and Ridge Regressors (Scikit-Learn, 2022a, 2022b).

The dataset is divided using 80% for the training and 20% for the test set. Five-fold cross-validation was used for both Random Forests and XGBoost to determine the models' hyperparameters.

4.2. Feature transformation

Before training the regressors with the dataset presented in Section 3, there is a need to transform the features into numerical variables so that the models can interpret them. There are three types of variables in the dataset: binary, nominal, and ordinal categorical variables. Table 6 links the 18 attributes described previously in Table 3 to these variable types.

Ordinal variables are encoded, assigning numbers to each category respecting the order of the attribute (e.g., the (0, 30] km/h speed limit bin is set to the number 0, and the (130, inf) bin, the number 6). On the other hand, both binary and nominal variables are transformed using the

One-Hot encoding technique, which creates a binary column for each observed category indicating its presence (Zheng & Casari, 2018). As a result, the 18 contextual attributes are converted to a 26-dimensional representation.

4.3. Feature importance and model interpretation

While machine learning techniques often outperform traditional actuarial models, their application in the insurance industry remains limited because of their difficulty in results interpretation (Henckaerts et al., 2021). This constraint is a principal requirement for insurance models, which need to be intuitively explainable to clients and regulators (European Parliament & Council of the European Union, 2016). Miller (2019) defines interpretability as "the level to which an observer is able to understand the cause of a model's choices". This paper aims to find such an understanding by looking at the relevance of each contextual feature on risk predictions.

A model-agnostic technique that allows examining feature importance is Shapley Additive Explanations (S. M. Lundberg & Lee, 2017; Molnar, 2022). This technique comes from cooperative game theory based on Shapley values and consists of computing the contribution of each feature to the model output. It allows explaining predictions of individual instances and a global model interpretation. Shapley values aim to fairly distribute the prediction among the feature set, which is obtained through an additive feature attribution method. The core idea

Trip 1										
Contextual variables							Risky events			Distance
speed limit, road type, road quality, road slope, ..., weather condition, wind speed							harsh braking, ..., speeding proportion			
(0, 30]	rural	(4, 6]	(-0.5, 0.5]	...	clear	[0, 10]	0.125	...	0.5	8
...										
(110, 130]	motorway	(0, 2]	(-0.5, 0.5]	...	clear	[0, 10]	0.2	...	0.2	10
Trip 2										
Contextual variables							Risky events			Distance
speed limit, road type, road quality, road slope, ..., weather condition, wind speed							harsh braking, ..., speeding proportion			
(0, 30]	rural	(4, 6]	(-0.5, 0.5]	...	clear	[0, 10]	0.2	...	0.35	10
...										
(110, 130]	motorway	(0, 2]	(-0.5, 0.5]	...	clear	[0, 10]	0.267	...	0.15	15
Context profiles aggregation										
Contextual variables							Risky events			Distance
speed limit, road type, road quality, road slope, ..., weather condition, wind speed							harsh braking, ..., speeding proportion			
(0, 30]	rural	(4, 6]	(-0.5, 0.5]	...	clear	[0, 10]	0.167	...	0.42	18
...										
(110, 130]	motorway	(0, 2]	(-0.5, 0.5]	...	clear	[0, 10]	0.24	...	0.17	25

Fig. 5. Example of the context profile aggregation using two trips. The contextual variables and risky events are simplified for clarity purposes. Units: speed limit [km/h], road quality [IRI], road slope [°], wind speed [km/h], distance [km].

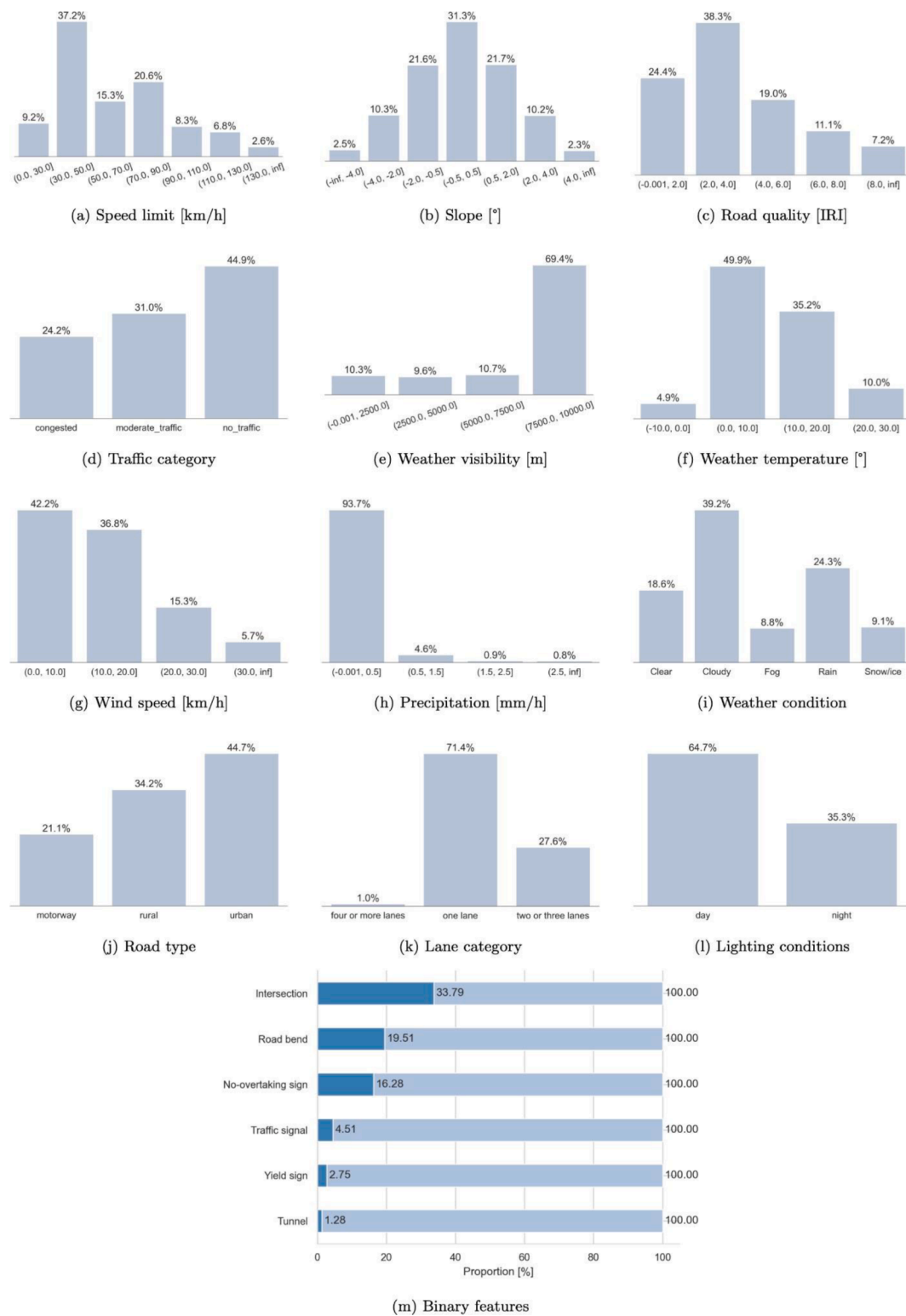


Fig. 6. Feature distribution of the features described in Table 3. Speed limits higher than 130 km/h refers to German roads with unlimited speed limits. Notes: Road quality is measured in the International Roughness Index (IRI) and temperature in degrees Celsius.

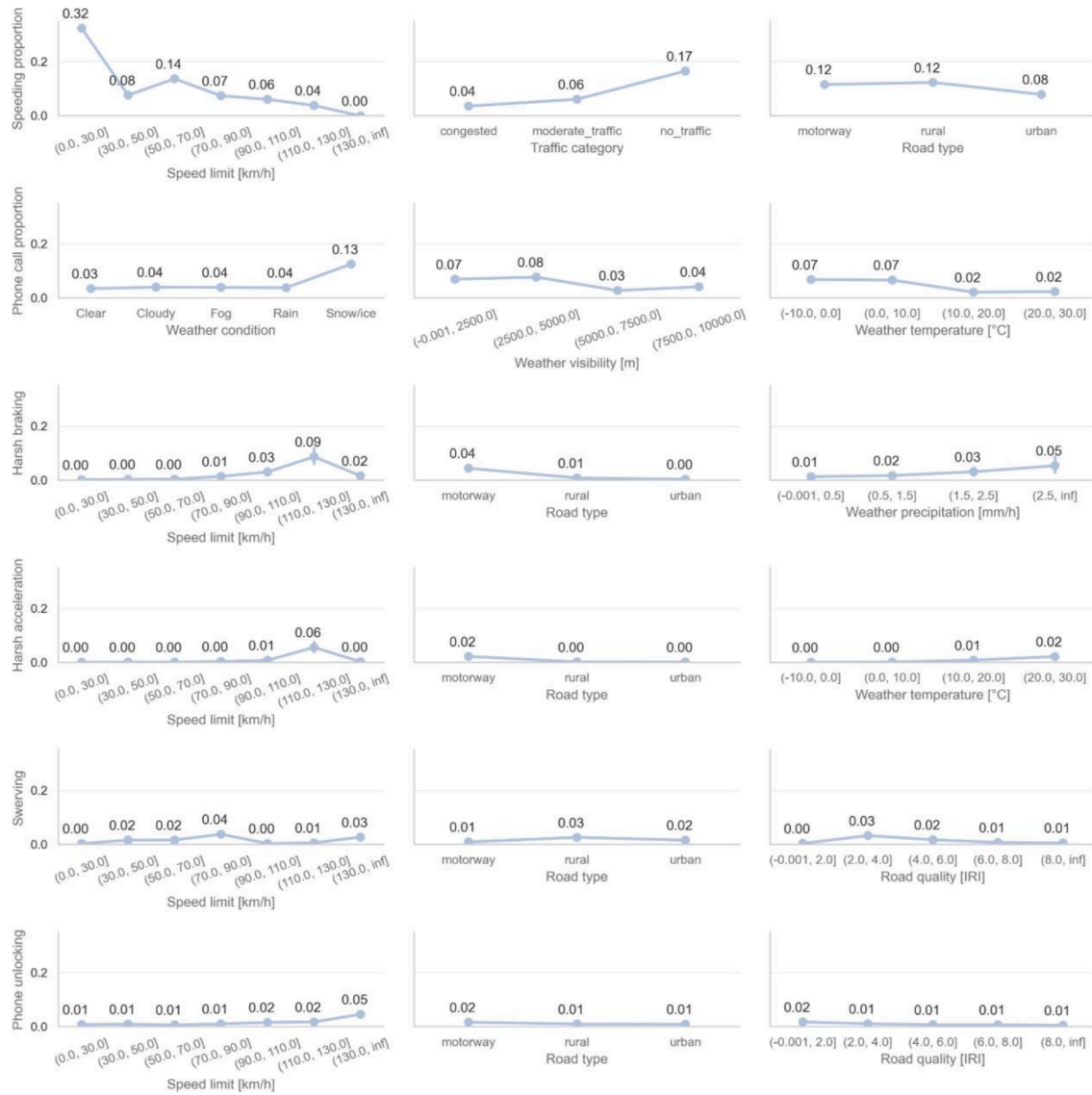


Fig. 7. Point-plots of the most relevant associations between features and risky events. Speed limits and road types present considerable correlations in most of them, followed by weather- and traffic-related variables. Speeding proportion is the event with the most pronounced correlation with both speed limits and traffic categories, being inversely proportional to an increase in speed limits and traffic jams.

Table 6

Links between variable types and attributes defined in Table 3.

Variable type	Attributes
Binary	Intersection, lighting condition, no-overtaking sign, road bend, traffic signal, tunnel, yield sign.
Nominal	Lane category, road type, weather condition.
Ordinal	Road quality, slope, speed limit, traffic category, weather precipitation, weather temperature, weather visibility, wind speed.

is that features with high absolute Shapley values are the most important. Equation 4 shows the global importance of each feature across the dataset, where m is the number of samples and $\phi_j^{(i)}$ is the Shapley value of feature j for the sample i .

$$Importance_j = \frac{1}{m} \sum_{i=1}^m |\phi_j^{(i)}|, \quad \forall j \in \text{Feature set} \quad (4)$$

Shapley values come from cooperative game theory (Shapley 1953). It is a method that allows obtaining a fair distribution of the “payout” among players based on each player’s contribution to the total

payout. A Shapley value is the average expected marginal contribution of one player after considering all possible combinations. In machine learning, it relates to the prediction task for an instance of the dataset where the players are the features of that instance. These features collaborate to obtain the prediction gain or cost, which is the difference between the prediction of that instance and the average prediction for all instances. The Shapley value is given by Equation 5, where S is the coalition of players (i.e., the subset of features used in the model) of the set N (of n players, i.e., the full set of features); v is a characteristic function that assigns values to subsets of players, and $v(S)$ describes the total expected sum of contributions that the players of S can obtain by cooperation. The sum covers all subsets S of N not containing the feature j . Thus, the Shapley value represents the contribution of feature j to the model prediction, weighted by all possible feature combinations.

$$\Phi_j(v) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(n - |S| - 1)!}{n!} [v(S \cup \{j\}) - v(S)] \quad (5)$$

Table 7

Model metrics by target events. XGboost presents the best performance in Root Mean Squared Error (RMSE) and Mean Poisson Deviance (MPD) for most events. The dummy regressor provides a benchmark to compare the models due to the observed variable imbalance (Section 3.3). The dataset is divided using 80% for the training and 20% for the test set. Five-fold cross-validation determined the models' hyperparameters. The models' criterion was selected according to their problem domain: Mean Squared Error (MSE) for ratio events and Poisson for count events.

Risky event	Dataset	Model sorted by best performance	RMSE	MPD
Cornering	Test set	XGBoost (objective: poisson; estimators: 300; max_delta_step: 0.8)	1.0250	0.1478
		RandomForest (criterion: poisson, estimators: 200)	1.0553	0.1748
		PoissonRegressor (fit_intercept: True)	1.1504	0.4300
		DummyRegressor (strategy: mean)	1.1505	0.4365
		MLPRegressor (n_hidden_units: (100, 150, 50))	2.3859	0.7568
	Training set	XGBoost (objective: poisson; estimators: 300; max_delta_step: 0.8)	0.7901	0.0828
		RandomForest (criterion: poisson, estimators: 200)	0.8180	0.1014
		PoissonRegressor (fit_intercept: True)	1.1560	0.4228
		DummyRegressor (strategy: mean)	1.1561	0.4298
		MLPRegressor (n_hidden_units: (100, 150, 50))	1.9590	0.6333
Harsh acceleration	Test set	XGBoost (objective: poisson; estimators: 50; max_delta_step: 1.0)	0.1288	0.0384
		RandomForest (criterion: poisson, estimators: 100)	0.1299	0.0408
		PoissonRegressor (fit_intercept: True)	0.1300	0.0585
		DummyRegressor (strategy: mean)	0.1300	0.0587
		MLPRegressor (n_hidden_units: (100, 100))	0.1588	0.0689
	Training set	XGBoost (objective: poisson; estimators: 50; max_delta_step: 1.0)	0.1272	0.0310
		RandomForest (criterion: poisson, estimators: 100)	0.1279	0.0321
		PoissonRegressor (fit_intercept: True)	0.1355	0.0710
		DummyRegressor (strategy: mean)	0.1355	0.0715
		MLPRegressor (n_hidden_units: (100, 100))	0.0994	0.0625
Harsh braking	Test set	XGBoost (objective: poisson; estimators: 50; max_delta_step: 1.0)	0.2841	0.0908
		RandomForest (criterion: poisson, estimators: 200)	0.2858	0.0911
		PoissonRegressor (fit_intercept: True)	0.2913	0.1433
		DummyRegressor (strategy: mean)	0.2913	0.1441
		MLPRegressor (n_hidden_units: (100, 150, 50))	0.3030	0.1776
	Training set	XGBoost (objective: poisson; estimators: 50; max_delta_step: 1.0)	0.2831	0.0798
		RandomForest (criterion: poisson, estimators: 200)	0.2728	0.0567
		PoissonRegressor (fit_intercept: True)	0.2938	0.1504
		DummyRegressor (strategy: mean)	0.2939	0.1516
		MLPRegressor (n_hidden_units: (100, 150, 50))	0.2896	0.1495
Phone unlocking	Test set	XGBoost (objective: poisson; estimators: 50; max_delta_step: 0.2)	0.2921	0.1343
		PoissonRegressor (fit_intercept: True)	0.2921	0.1354
		DummyRegressor (strategy: mean)	0.2922	0.1361
		RandomForest (criterion: poisson, estimators: 200)	0.2927	0.1347
		MLPRegressor (n_hidden_units: (100, 100))	0.3149	0.1559
	Training set	XGBoost (objective: poisson; estimators: 50; max_delta_step: 0.2)	0.2802	0.1201
		PoissonRegressor (fit_intercept: True)	0.2807	0.1254
		DummyRegressor (strategy: mean)	0.2808	0.1260
		RandomForest (criterion: poisson, estimators: 200)	0.2720	0.0764
		MLPRegressor (n_hidden_units: (100, 100))	0.2293	0.1262
Phone call proportion	Test set	RandomForest (criterion: mse, estimators: 400)	0.0697	–
		XGBoost (objective: mse, estimators: 1300)	0.0775	–
		MLPRegressor (n_hidden_units: (100, 100))	0.0953	–
		RidgeRegressor (fit_intercept: False)	0.1129	–
		DummyRegressor (strategy: mean)	0.1147	–
	Training set	RandomForest (criterion: mse, estimators: 400)	0.0253	–
		XGBoost (objective: mse, estimators: 1300)	0.0525	–
		MLPRegressor (n_hidden_units: (100, 100))	0.0828	–
		RidgeRegressor (fit_intercept: False)	0.1072	–
		DummyRegressor (strategy: mean)	0.1103	–
Speeding proportion	Test set	RandomForest (criterion: mse, estimators: 300)	0.1294	–
		MLPRegressor (n_hidden_units: (100, 100))	0.1614	–
		RidgeRegressor (fit_intercept: False)	0.1831	–
		DummyRegressor (strategy: mean)	0.2102	–
		XGBoost (objective: mse, estimators: 1200)	0.0833	–
	Training set	RandomForest (criterion: mse, estimators: 300)	0.0463	–
		MLPRegressor (n_hidden_units: (100, 100))	0.1425	–
		RidgeRegressor (fit_intercept: False)	0.1774	–
		DummyRegressor (strategy: mean)	0.2073	–

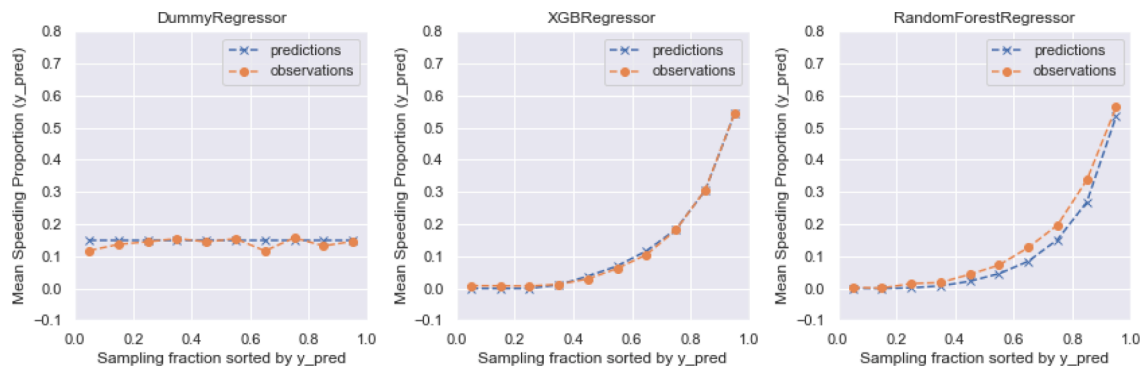


Fig. 8. Comparison between predicted and observed speeding proportions in the test set. The test set is sorted according to the predictions in ascending order and then divided into ten ranges. XGBoost is the regressor that shows the best consistency between predicted and observed values, aligned with the model metrics and the mean predicted speeding proportion (predicted 14.19%, actual 14.05%). XGBoost is followed by Random Forest, which underestimates the speeding proportion of some contexts. Both methods are compared with a dummy regressor that always returns the mean observed in the training set.

5. Results

5.1. Performance metrics per target event

Table 7 shows the model results for the six target events in the test and training set, sorted by the best performance in the test set. The performance for all models is measured through the root mean squared error (RMSE) with weights according to the distance driven in each context so that predictions of frequent contexts are more relevant to the score. The RMSE measures the standard deviation of the prediction errors, where low values are associated with better models. Since count events are assumed to follow a Poisson distribution, they also include the mean Poisson deviance (MPD), showing consistent results with RMSE. MPD allows a better distinction among Poisson models, where RMSE may be susceptible to class imbalances (e.g., there is a difference of 0.0012 between the best and worst RMSE for harsh acceleration, whereas, with MPD, the difference is 0.0203). The table includes a dummy regressor that returns the mean observed value in the training set as a metric reference due to the imbalance toward no dangerous events. The results show that XGBoost slightly outperforms Random Forest in most situations, except when modelling phone call proportions. Particularly, ratio events are associated with the best RMSE scores, followed by harsh acceleration and braking. For these events, Random Forest seems to overfit the training set; for example, in modelling speeding proportion, its RMSE is 0.0463 and 0.1294 for the training and test set, respectively.

Fig. 8 compares predicted and observed values for speeding proportion models in the test set. The dataset is sorted according to the predictions in ascending order and then divided into ten ranges (e.g., the first range has the first 10% of the test set, and the tenth range has from the 90th percentile up to the last sample). XGBoost, capped between 0 and 1, shows the best consistency between predicted and observed values, which is aligned with the model metrics. The overall predicted value of speeding was 14.19%, similar to the actual 14.05%. XGBoost is followed by Random Forest, which underestimates the speeding proportion of some contexts. Both methods are compared with a dummy regressor that returns the mean observed in the training set. The remaining models experience similar results and are included in Appendix A.

5.2. Feature importance and model interpretation

The models mentioned above allow the analysis of the most relevant features that influence their predictions, ranking the feature set according to their importance. Such rankings are illustrated in Fig. 9 through SHAP summary plots where the most relevant features appear at the top. The features are sorted by the mean of absolute SHAP

magnitudes over all samples in the test set, according to Equation 4 and detailed in Appendix B. The core idea is to represent feature importance through the feature impact on the model output. Each point in the x-axis represents a Shapley value for an instance of the dataset, coloured according to the feature value, from low (blue) to high (red). The Shapley value is proportional to the feature influence on the prediction, where positive values increase model predictions and negative ones decrease the prediction. It is worth noting that the illustrated effects refer to the correlations using marginal effects and do not necessarily indicate a causal effect, which needs to be investigated in future works (Wen et al., 2022).

Following an analysis of the rankings presented in Fig. 9, speed limit, weather temperature, and wind speed appear in the top ten most important features for all events. Remarkably, the speed limit attribute is in the top five of all rankings. These contextual features are followed by road slope and traffic category, appearing in the top ten of all events except harsh braking. Traffic signs and tunnels do not seem to contribute to the predictive power of these models, most likely due to their relatively low frequency in the dataset and the models' bias according to distance travelled.

Table 8 details the observed relationships between risky events and contextual factors from Fig. 9. The results show that traffic congestion, urban roads, poor road quality, and curves lower the predicted value of speeding. Low-speed limits and daylight conditions mainly influence increases in this event; pronounced negative slopes also seem to be generally associated with speeding, although with a less clear relationship. These observations align with previous studies on speeding analysis and Fig. 7 (Bärgman et al., 2017; Jun et al., 2011). A slight effect of high temperatures on increasing the predictive value of speeding is also observed, which is a psychological aspect that aligns with the pilot study of Chowdhury (2015). Harsh manoeuvre predictions (i.e., near-misses) are mainly influenced by road types, layout, and weather conditions. In particular, high levels of precipitation, motorway usage, curves, and tunnels tend to increase the model outputs for harsh acceleration and braking. When approaching intersections, harsh braking predictions generally rise, indicating evasive manoeuvres for rear-end accidents. Cornering events are positively influenced by curves, intersections, and one-lane roads and negatively affected by free-flow traffic conditions and tunnels. These three harsh events tend to decrease with low temperatures, and high-speed limits are inversely proportional to harsh acceleration predictions.

As for distraction events, late driving and adverse weather conditions like snow correspond to higher engagement in phone calls, and moderate to high temperatures decrease such engagement. Manual interactions with the mobile phone are shown to increase while approaching intersections, which is a factor found to reduce reaction distance (Choudhary and Velaga, 2019). Other features positively

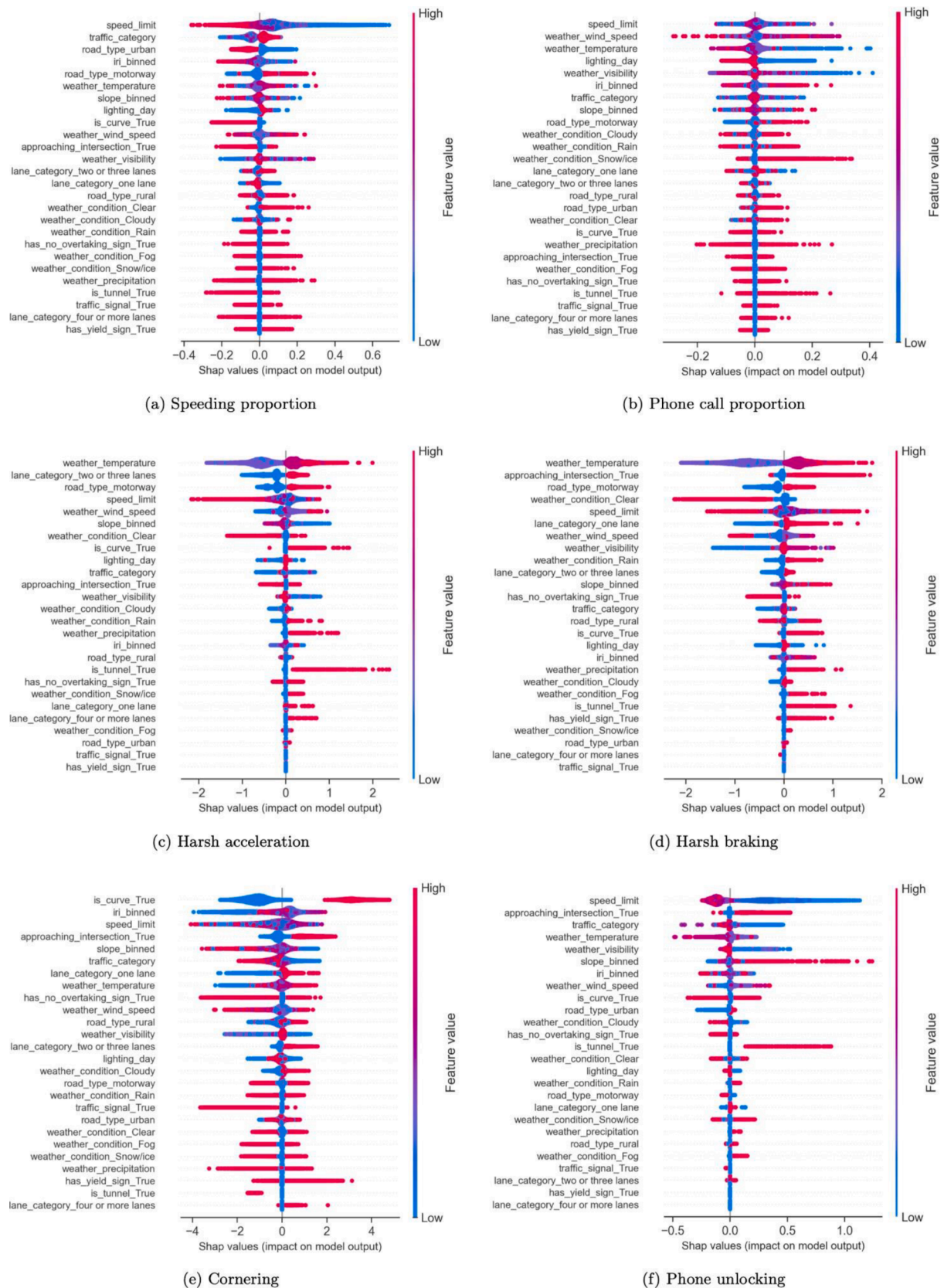


Fig. 9. Feature importance for the six risky events using SHAP summary plots. The features are sorted according to their importance, with the most relevant features at the top. The values on the x-axis indicate a Shapley value for instances of the test set (overlapping values are stacked along the y-axis), and the colour scale represents the feature value from low to high.

Table 8

Contextual factors influencing the predictions of risky events. The relationships are obtained from the SHAP summary plots in Fig. 9. The factors are sorted by decreasing feature importance, skipping the ones with negligible impact on the model output or unclear relationships.

Risky event	Factors that influence the prediction	
	Positive influence	Negative influence
Cornering	Road bend, intersection, one-lane road	Free-flow traffic, good road quality, low temperature, tunnel
Harsh acceleration	Motorway, high wind speed, negative slope, road bend, rain, tunnel	Low temperature, high-speed limit
Harsh braking	Intersection, motorway, one-lane road, rain, positive slope, road bend, poor road quality, tunnel	Low temperature, clear weather conditions
Phone call proportion	Nighttime driving, motorway, snow/ice conditions	Moderate to high temperature
Phone unlocking	Low-speed limit, intersection, traffic congestion, positive slope, tunnels, nighttime driving, rain.	Non-urban road, motorway
Speeding proportion	Low-speed limit, daylight driving, negative slope	Traffic congestion, urban roads, poor road quality, road bend

related to phone unlocking are low-speed limits, traffic jams, driving in tunnels, and nighttime driving. An interesting observation is that pronounced positive slopes and rain also increase the predicted density of phone unlocking. However, these results should be taken with care because of the low predictive performance of this risky event reflected in an MPD slightly better than the dummy regressor. As observed in Fig. 7, there is a lack of a precise contextual predictor for this event which might indicate that the propensity to manually interact with the phone is rather more dependent on driver behaviour than the driving context (Hossain et al., 2022).

SHAP also allows understanding of the dependencies between features and their collective impact on the model output. Fig. 10 shows the effects of some of the top-ranked contextual features for the six target events. The scatter plots show, on the y-axis, the distribution of the SHAP values for a given top-ranked contextual attribute, whereas the x-axis shows the respective value. The colour variable represents another top-ranked feature that might interact with the attribute on the x-axis. Such dependence plots present the dispersion in the feature's contribution to the model's output given the feature value.

Dependence plots illustrate potential interaction effects through vertical colour patterns. For example, Fig. 10 shows that high-speed limits tend to increase speeding predictions at warm temperatures (more than 20 °C). Driving on high-speed roads in warm temperatures increases the likelihood of having harsh acceleration and braking, while the inverse effect is observed in low temperatures. This observation aligns with a similar finding by Zhu et al. (2017), who found that risky drivers tend to demonstrate aggressive manoeuvres on freeways. The effect of the road quality on cornering effects is non-linear, where most of the increases in cornering predictions happen in moderate road quality roads. This observation can be expected since a low IRI is typically associated with high-speed roads. In contrast, extremely high values of IRI often refer to low-speed and infrequent roads that require more caution from the driver. Such a finding aligns with the observations of Montella & Imbriani (2015), who posited that inconsistencies in road quality increase driving risk. Another non-linear pattern is observed in phone unlocking events where the effects of speed limit tend to be negative after 50 km/h with exceptions in speed limits between 90 and 100 km/h, where free-flow circumstances tend to increase the likelihood of distraction.

As stated in Section 2.5, SHAP is often-used to extract interpretations of the relationships learned by complex machine learning models and understand different risk-level predictions. Fig. 11 applies SHAP's force

plots to illustrate the feature contributions of three harsh braking predictions on the test set having different risk levels: high-risk, average-risk, and low-risk. The charts show the contribution of each contextual variable to the model output $f(x)$, which can be compared against the average prediction (i.e., the base value). Each arrow represents the feature's positive (red) or negative (blue) contribution to the model prediction, proportional to its width, and before applying the Poisson link function used by XGBoost.

The force plot at the top of Fig. 11 shows a high-risk prediction, where exposure to intersections, curves, and limited weather visibility shift the model prediction to high values. Drivers frequently exposed to those driving contexts present more exposure to harsh braking events. These harsh manoeuvres are associated with rear-end accidents (Kusano & Gabler, 2012). Therefore, driving under these conditions means increased exposure to accidents and a higher driving risk than those exposed to a safer driving context like the one at the bottom in Fig. 11. In such a case, low weather temperatures, clear weather conditions, moderate speed limits on motorways lower the likelihood of harsh braking events, which is aligned with the dependence plot of Fig. 10 and the summary plot of Fig. 9.

6. Conclusions

This research investigates the importance of a contextual layer of information for driving risk assessment. Through a naturalistic driving dataset comprising around 79,000 km from 32 anonymous drivers, the article presents a novel analytics methodology to determine the predictive significance of driving context combinations for near-misses, speeding, and distraction events. Furthermore, the most important contextual factors are identified and ranked. The results show that the driving context profile carries a significant predictive power for near-misses, speeding, and distraction events. A correlation between risky events and road accidents has been identified in past research. Therefore, driving under contextual conditions that increase the likelihood of these risky events augments the exposure to accidents and, hence, the driving risk.

To the best of the authors' knowledge, this is the first paper that applies Shapley Additive Explanations (SHAP) from a risk perspective based on near-misses, speeding, distraction events and a comprehensive set of contextual attributes coming from naturalistic driving data. The results show that the features that carry the most significant power for predicting near-misses, speeding, and distraction events are speed limit, weather temperature, wind speed, traffic conditions and road slope, appearing in the top ten relevant contexts of at least five of the studied risky events. They are followed by road infrastructure factors like road bend, quality, and type, along with weather visibility. The paper analysed feature effects on model outputs and found that high-speed limits combined with free-flow conditions are associated with increases in phone unlocking events. As expected, high-speed limits are linked with decreases in harsh acceleration events, and traffic congestion increases the density of predicted cornering and phone unlocking events. Several observations are found for weather and lighting conditions: (i) Warm temperatures and high-speed limits tend to increase the speeding likelihood; (ii) low temperatures and high-speed roads tend to decrease the density of harsh acceleration and braking; (iii) the usage of motorways at temperatures higher than 10 °C increases the predicted frequency of harsh braking; (iv) rainy conditions increase the predicted value of harsh acceleration, braking and distraction events; (v) daylight driving is associated with increased speeding and nighttime driving with distraction events. Regarding road layout attributes, the road quality has a non-linear relationship with cornering events where good road quality (i.e., consistent pavement) decreases cornering likelihood. In contrast, poor road quality tends to decrease speeding and increase harsh braking and cornering. Exposure to intersections increases harsh braking, cornering, and phone unlocking.

The contributions of this paper are relevant for insurers and road

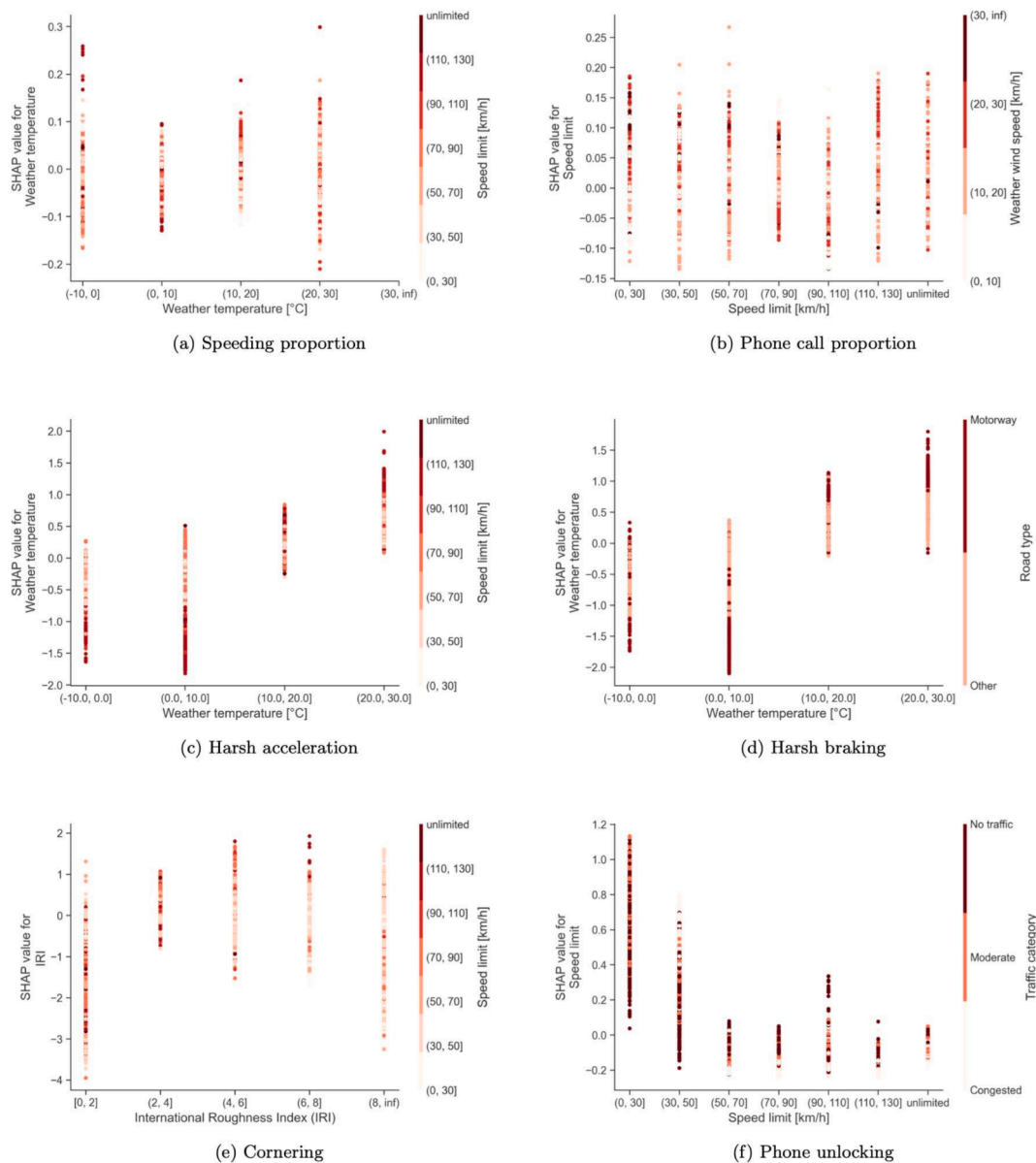


Fig. 10. SHAP dependence plots for top-ranked contextual features in the six target events. The scatter plots show, on the y-axis, the distribution of the SHAP values for a given contextual feature. The x-axis presents the respective feature value. Each point is a prediction on the test set where the colour represents the magnitude of the variable on the right.

safety stakeholders willing to develop or incorporate contextual information into their risk and pricing models through a tool that allows model outputs interpretations. They represent the first step toward Pay-Where-You-Drive (PWYD) schemes based on driving context profiles. Such schemes might leverage driver-centric contextual ratings and their exposure to accidents to enhance the predictive power of traditional actuarial models. PWYD introduces an alternative approach for risk assessment that reduces the privacy concerns typically associated with UBI schemes collecting and storing high-frequency geo-location data. In these schemes, the policyholders' risk is determined by understanding their driving context profile through questionnaires or low-sampling rate data collection. While this research showed the relevance of the driving context on the driving risk, quantifying the expected change in the insurance premium would require an extended analysis using historical claims data instead of near-misses. The use of contextual

information also helps road safety stakeholders to derive policies to mitigate hazardous situations. In particular, the application of the feature importance based on a comprehensive set of contextual attributes contributes to an understanding of the dependencies among risk factors related to near-misses, speeding and distraction events.

Understanding driving context profiles becomes essential for the next generation of risk assessment models targeting higher levels of vehicular autonomy, where the human-related factors are reduced, and the driving context constrains the performance of the vehicle actuators (Sheehan et al., 2017). These models could leverage naturalistic driving data to detect dangerous contextual hotspots for automated vehicles or Advanced Driver Assistance Systems (ADAS), similarly to the behavioural hotspots detection of Ryan et al. (2020). This research contributes a novel dataset for future research on the relationship between driving context and risk and its added value to risk models, fostering innovation

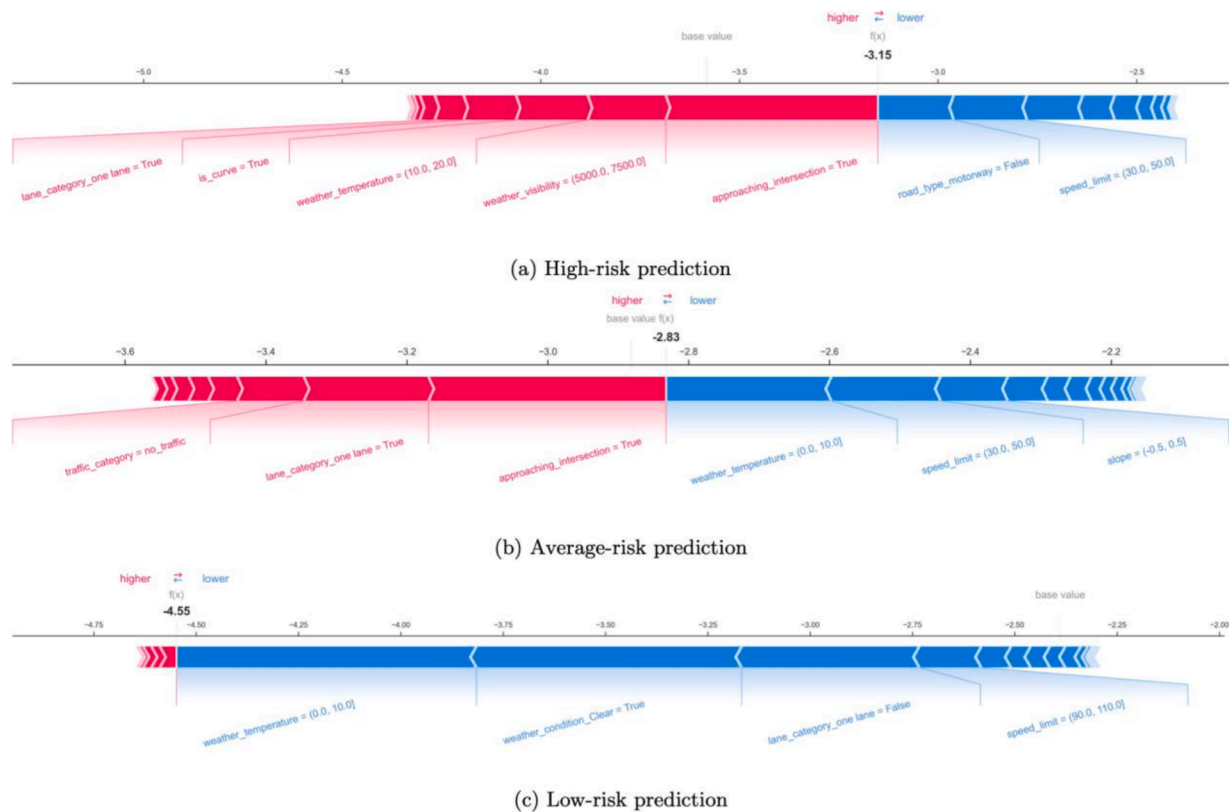


Fig. 11. Force plots for harsh braking predictions of three test set instances. Top – high-risk prediction: predicted event density above average; medium – average-risk prediction: predicted event density close to the base value; bottom – low-risk prediction: predicted event density below the average. The charts show the contribution of each contextual variable to the model output $f(x)$. The model output can be compared against the base value, which refers to the average prediction. Each arrow represents the feature's positive (red) or negative (blue) contribution to the model prediction, proportional to its width. The values are contributions before applying the Poisson link function used by XGBoost. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

in the motor insurance industry.

There are some lines of thought for future research extending upon the contribution of this paper. The introduction of multi-output regression for correlated target events, such as harsh acceleration and braking, is worth exploring. At the moment of writing this manuscript, some of the techniques used in this paper do not support multi-output regression, jointly modelling target variables, using the Poisson objective or criterion. Performance metrics are also constrained. However, alternatives based on Sklearn's MultiOutputRegressor (Scikit-Learn, 2022c) might be developed, or other techniques, such as Gaussian Processes, might be further explored. Similarly, deep learning techniques present opportunities to further study the predictive power of the driving context through SHAP's DeepExplainer (Lundberg, 2023). The lack of information about the vehicle information and driver demographic, including driving history, constitutes a limitation for this study since the exposure to risky events may differ by driver group. Hence, more detailed studies may control for these variables to identify the impact of the driving context on different driver populations. Future work might explore the causal relationships between contextual features and dangerous events using controlled experiments, complementing the findings presented in Section 5.2, which are based on correlations using marginal effects on model predictions.

CRediT authorship contribution statement

Leandro Masello: Conceptualization, Methodology, Software, Investigation, Data curation, Writing – original draft, Visualization. **German Castignani:** Conceptualization, Methodology, Investigation, Writing – review & editing, Supervision. **Barry Sheehan:** Conceptualization, Methodology, Investigation, Writing – review & editing, Supervision. **Montserrat Guillen:** Conceptualization, Writing – review & editing, Supervision. **Finbarr Murphy:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Author affiliations with Motion-S: Leandro Masello and German Castignani

Data availability

The authors have made the dataset publicly available under the Creative Commons Attribution-NonCommercial 4.0 International licence (Creative Commons, 2020)

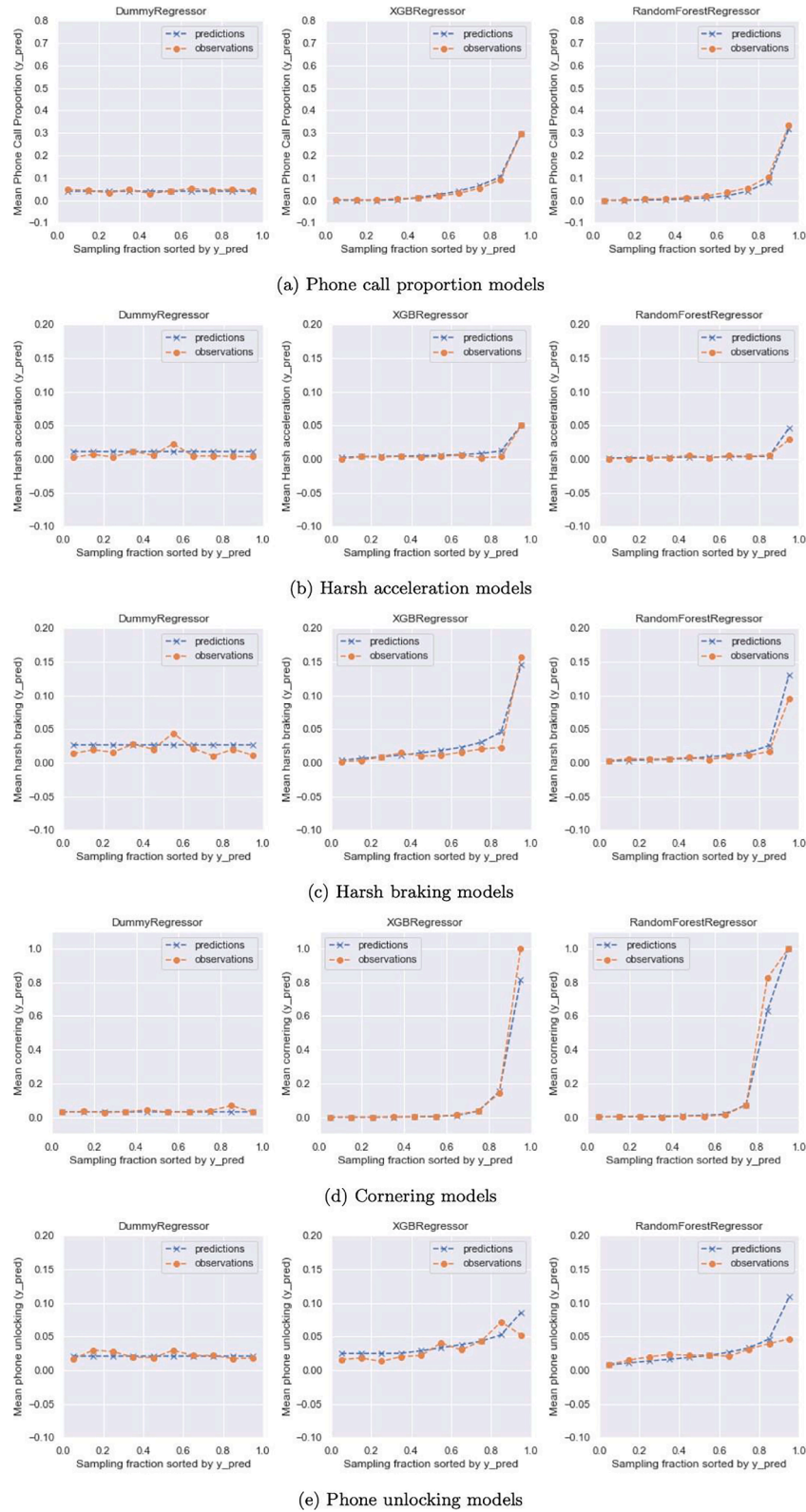


Fig. A1. Complementary to Fig. 8. Comparison of the modelling techniques for phone call proportion, harsh acceleration, harsh braking, cornering, and phone unlocking. The scaling of phone call proportions is the same as in Fig. 8 since they are both ratio events. Count events have the same scale, except for cornering events which, due to some outliers, are capped at 1.

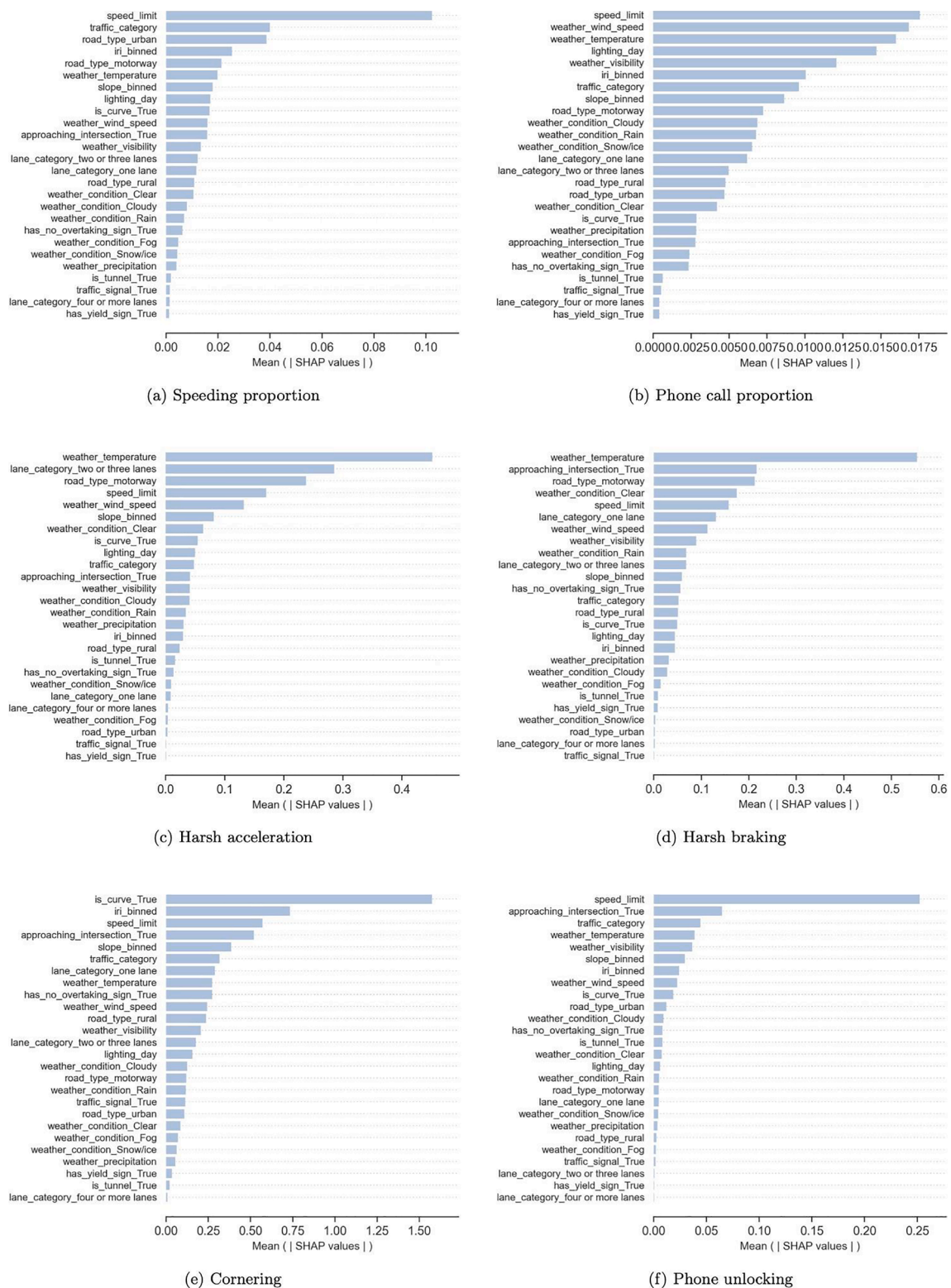


Fig. A2. Complementary to Fig. 9. Feature importance using SHAP bar plots. The subplots show the mean of the absolute SHAP values for each feature for the six studied events. Higher values represent higher feature importance. The y-axis order in each subplot aligns with the one in Fig. 9.

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Appendix A

Appendix B

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